Vectors

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ **Contents & learning path**

1. A vector has size and direction
2. Addition and relative motion
3. The dot product: projection and flux
4. The cross product: rotation and orientation
5. Unit vectors and coordinate choices
6. Physical checks
7. Lecture + Problems: predict, calculate, check
8. Further practice and provenance

1. A vector has size and direction

Scalars such as mass and temperature are fully described by a number and a unit. Vectors such as displacement, velocity, force and angular momentum also need a direction. In two dimensions,

$$\mathbf{A} = (A_x, A_y), \qquad |\mathbf{A}| = \sqrt{A_x^2 + A_y^2}. \tag{1}$$

If a vector has magnitude A and angle θ from the x-axis, then $A_x = A\cos\theta$ and $A_y = A\sin\theta$. Components are not extra physics; they are a bookkeeping system that makes addition easy.

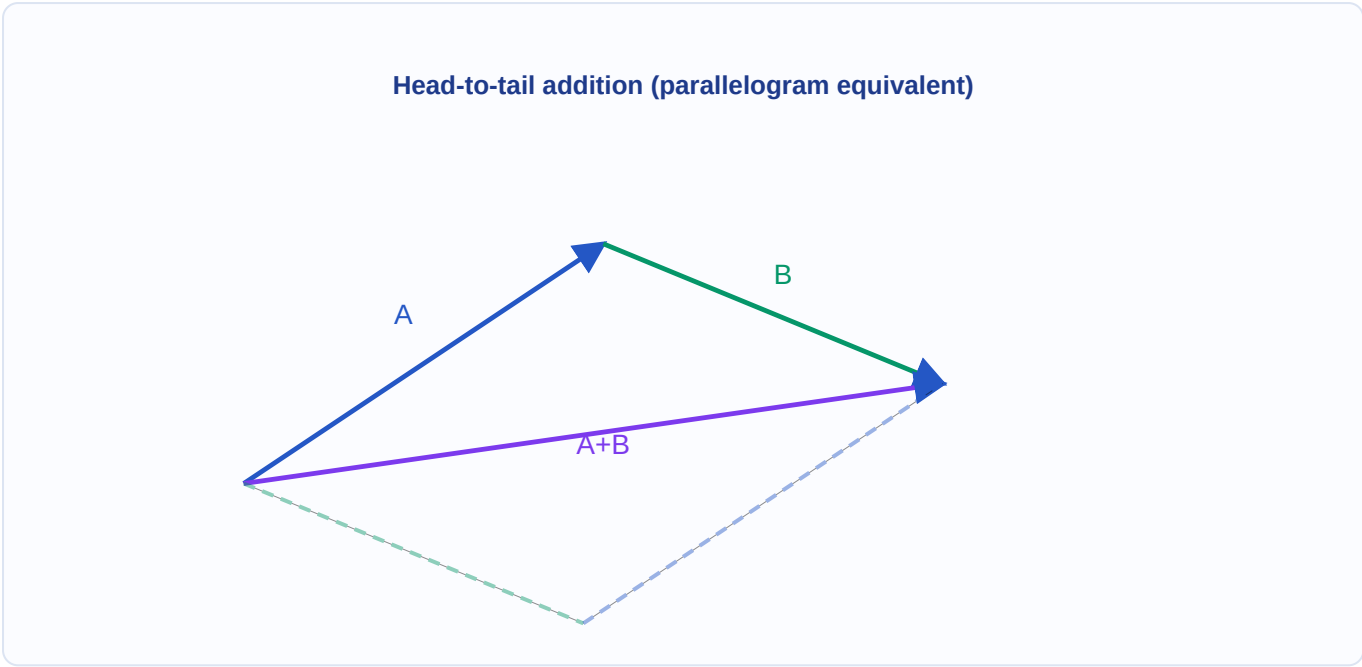


Figure 1. Components make the same rule algebraic: add x with x and y with y.

2. Addition and relative motion

Vectors add head-to-tail, or component by component:

$$\mathbf{A} + \mathbf{B} = (A_x + B_x,\ A_y + B_y). \tag{2}$$

Relative velocity is a vector difference: $\mathbf{v}_{A/B} = \mathbf{v}_A - \mathbf{v}_B$. This is why a star's proper motion, Earth's motion and the observed radial velocity must be treated as directions, not just signed numbers.

3. The dot product: projection and flux

The dot product is

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y = |\mathbf{A}||\mathbf{B}| \cos\theta. \tag{3}$$

It measures how much one vector points along another. For a uniform field $\mathbf{E}$ through a flat oriented surface with area vector $\mathbf{A} = A\hat{\mathbf{n}}$, the signed flux is $\Phi = \mathbf{E} \cdot \mathbf{A} = EA\cos\theta$, with units of field times area. It is zero when the field lies in the surface (perpendicular to the normal), positive when it points along the chosen normal, and negative when it points opposite.

4. The cross product: rotation and orientation

In three dimensions,

$$|\mathbf{A}\times\mathbf{B}| = |\mathbf{A}||\mathbf{B}|\sin\theta, \tag{4}$$

with direction given by the right-hand rule. Torque is $\boldsymbol{\tau} = \mathbf{r}\times\mathbf{F}$; angular momentum is $\mathbf{L} = \mathbf{r}\times\mathbf{p}$. In two dimensions, the cross product has only a z-component: $A_xB_y - A_yB_x$. Its sign tells whether the rotation from $\mathbf{A}$ to $\mathbf{B}$ is counterclockwise or clockwise.

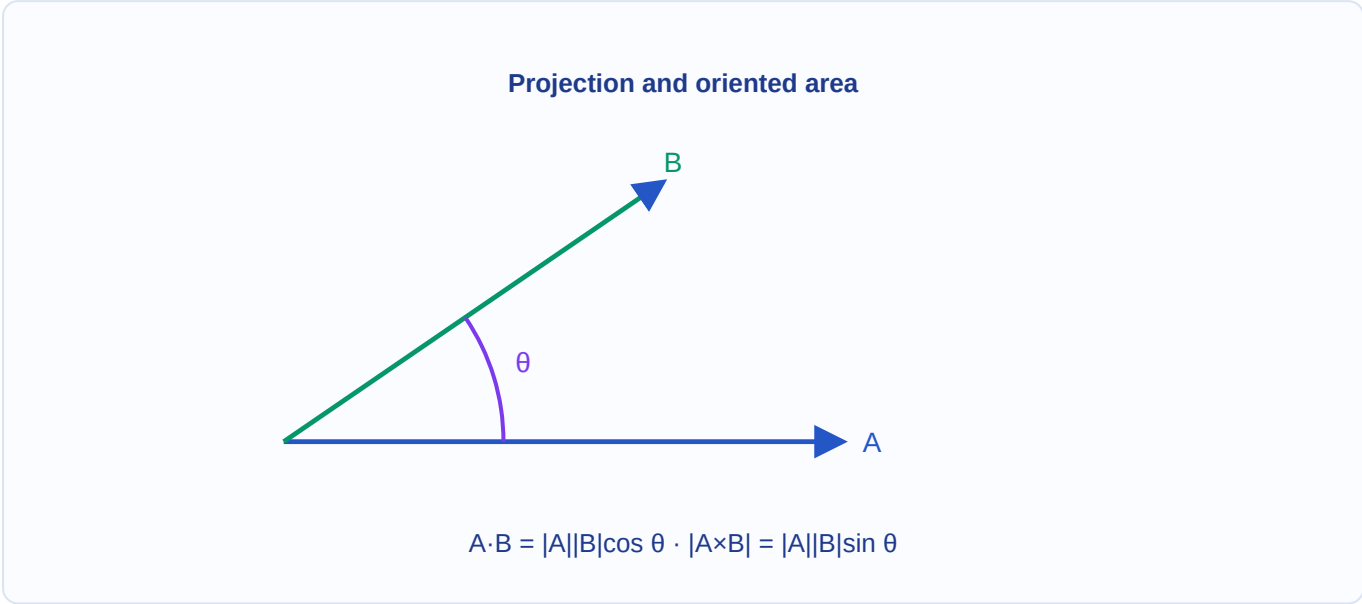


Figure 2. Flux uses a dot product; angular momentum and torque use cross products.

5. Unit vectors and coordinate choices

A unit vector has magnitude 1 and specifies direction: for $\mathbf{A}\neq\mathbf{0}$, $\hat{\mathbf{A}} = \mathbf{A}/|\mathbf{A}|$. The zero vector has no direction. Choosing axes wisely can make a problem trivial. Align one axis with a velocity, a force or a line of sight. In spherical astronomy, directions are often written with right ascension and declination, but local calculations may be easier in horizontal or Cartesian components.

6. Physical checks

Vector equations must be dimensionally consistent. A dot product of two velocities has units of velocity squared; a cross product of position and momentum has units of angular momentum. If the answer is a vector, state its magnitude and direction or its components. If the answer is zero, ask whether the vectors are perpendicular, parallel, or equal and opposite.

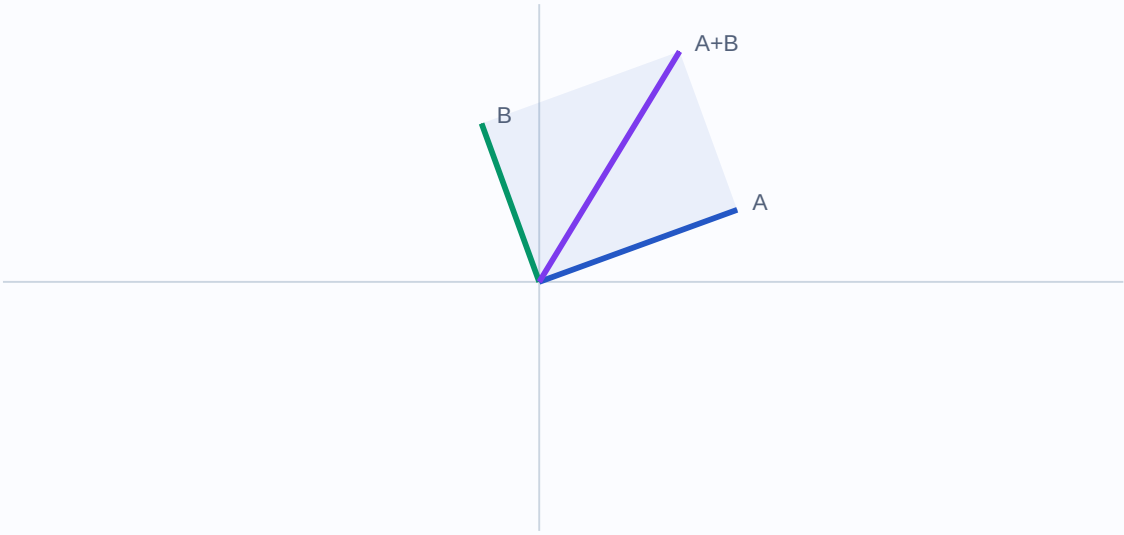
INTERACTIVE EXPERIMENT

Add, project and rotate in two dimensions.

The parallelogram shows A+B. The dot product measures projection; the z-component of the cross product measures oriented area.

A+B	|A+B|	A·B
(3.33, 5.47)	**6.40**	**0.00**

(A×B)z	Angle between
20.00	**90.0°**



Dot product = |A||B|cos θ; cross-product z component = |A||B|sin θ. The parallelogram diagonal is the vector sum.

Try this. Make the vectors perpendicular, then parallel. What happens to the dot and cross products?

Snapshot of the experiment. Open the HTML edition to change the controls.

A reliable method. Draw the vectors. Choose axes. Resolve into components. Add or multiply with the correct rule. Convert back to magnitude and direction only at the end.

7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — Components. A vector has magnitude 10 and points 30° above the x-axis. Find its components.

Worked solution →

[1] Problem 2 — Vector sum. Add $\mathbf{A} = (3, 4)$ and $\mathbf{B} = (-1, 2)$. Find the magnitude and direction of the sum.

Worked solution →

[1] Problem 3 — Dot product. Find $\mathbf{A} \cdot \mathbf{B}$ for $\mathbf{A} = (2, 3)$ and $\mathbf{B} = (4, -1)$. Is the angle between them acute, right or obtuse?

Worked solution →

[2] Problem 4 — Angle between vectors. Find the angle between $\mathbf{A} = (1, 0)$ and $\mathbf{B} = (1, \sqrt{3})$.

Worked solution →

[2] Problem 5 — Cross product. Find the z-component of $\mathbf{A} \times \mathbf{B}$ for $\mathbf{A} = (3, 0)$ and $\mathbf{B} = (0, 4)$. What is the direction?

Worked solution →

[2] Problem 6 — Relative velocity. Star A moves at $(20, 10)$ km/s and star B at $(5, 25)$ km/s in the same frame. Find A's velocity relative to B and its speed.

Worked solution →

[2] Problem 7 — Flux through a surface. A uniform field of magnitude 5 passes through a surface of area 2 whose normal makes 60° with the field. Find the flux.

Worked solution →

[2] Problem 8 — Torque. A force $\mathbf{F} = (0, 10)$ N acts at position $\mathbf{r} = (2, 0)$ m. Find the torque about the origin.

Worked solution →

[3] Problem 9 — Projection. Project $\mathbf{A} = (4, 2)$ onto $\mathbf{B} = (1, 1)$. Give the vector projection and its length.

Worked solution →

[3] Problem 10 — Angular momentum direction. A planet moves counterclockwise in the xy-plane around the Sun at the origin. At $\mathbf{r} = (1, 0)$ AU its velocity is $(0, v)$. What is the direction of $\mathbf{L} = \mathbf{r} \times m\mathbf{v}$? Does it change for a closed Keplerian orbit?

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.